

The empirical recurrence for OEIS A295118: recovering a conjecture that is not written down

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Abstract

OEIS A295118 records “Empirical recurrence of order 78”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 82$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 78, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 78 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A295118 is “Number of $n \times 6$ 0..1 arrays with each 1 horizontally or vertically adjacent to 1 or 4 1s.”. It has offset 1 and begins

9, 85, 665, 5793, 49302, 420519, 3590821, 30650456, ...

The entry states:

Empirical recurrence of order 78 (see link above)

As of the “Last modified” line on the live entry (Nov 15 2017 06:20:07, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4096$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 82$ of the 4096, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 82$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 164$ exact terms of the model returns order 78 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{78} c_i a(n-i),$$

c_1	3	c_{21}	−9708511	c_{41}	1012755844	c_{61}	400477
c_2	26	c_{22}	12799189	c_{42}	−234801094	c_{62}	−2365975
c_3	145	c_{23}	−8027488	c_{43}	−474412415	c_{63}	91641
c_4	335	c_{24}	−7257001	c_{44}	832341435	c_{64}	−779343
c_5	308	c_{25}	−7287417	c_{45}	−917252265	c_{65}	−67885
c_6	−2850	c_{26}	−16827866	c_{46}	909212295	c_{66}	119469
c_7	−14868	c_{27}	38358768	c_{47}	−654504014	c_{67}	106120
c_8	−6518	c_{28}	1713834	c_{48}	442222470	c_{68}	62547
c_9	105921	c_{29}	−87775469	c_{49}	−207885094	c_{69}	3822
c_{10}	101336	c_{30}	47347560	c_{50}	146300377	c_{70}	−910
c_{11}	−416378	c_{31}	−81897056	c_{51}	−9954718	c_{71}	−4452
c_{12}	−475052	c_{32}	−183297632	c_{52}	−8257437	c_{72}	−13
c_{13}	863402	c_{33}	763878572	c_{53}	27741094	c_{73}	419
c_{14}	1499441	c_{34}	−1563029287	c_{54}	−26371825	c_{74}	218
c_{15}	−569161	c_{35}	2272961393	c_{55}	8673182	c_{75}	−28
c_{16}	−2606566	c_{36}	−2747105322	c_{56}	−20416503	c_{76}	−25
c_{17}	−1907319	c_{37}	3239222929	c_{57}	3406875	c_{77}	−3
c_{18}	2762634	c_{38}	−3271602041	c_{58}	−17896257	c_{78}	1
c_{19}	8295526	c_{39}	2699753761	c_{59}	−1457707		
c_{20}	−4658770	c_{40}	−1911075871	c_{60}	−8632298		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 78, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $78 + 4$ terms removed returns order 78 as well, so $\deg q_{\min} = 78 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $78 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 78 that this sequence satisfies — and that family turns out to have exactly one member.

References

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